Intro and First Day Stuff

Lecture 1 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

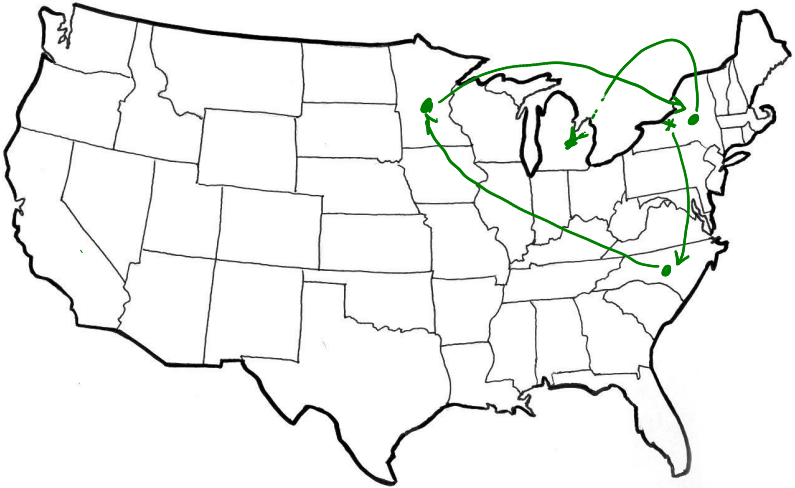
Mon, Jan 10, 2022

People in this lecture

Dr. Munch (she/her)
Depts of CMSE and Math

Emily Bolger (she/her)
Graduate Student, CMSE, MSU

About me

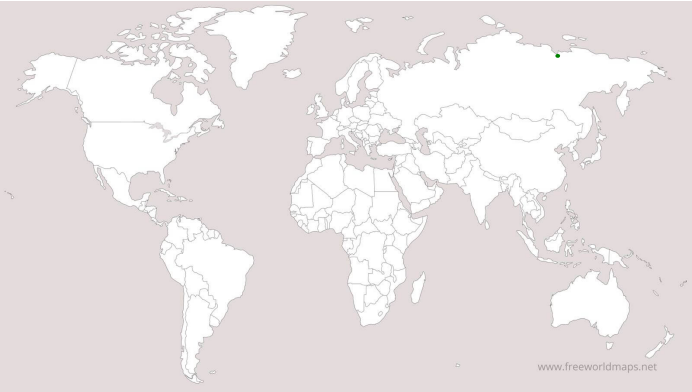


About you..... Annotate me!

In Michigan?

Not

Annotate



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Waterproof Paper.com

What is this course about?

Topics:

- Fundamental concepts of data science
- Regression
- Classification
- Dimension reduction
- Resampling methods
- Tree-based methods, etc.

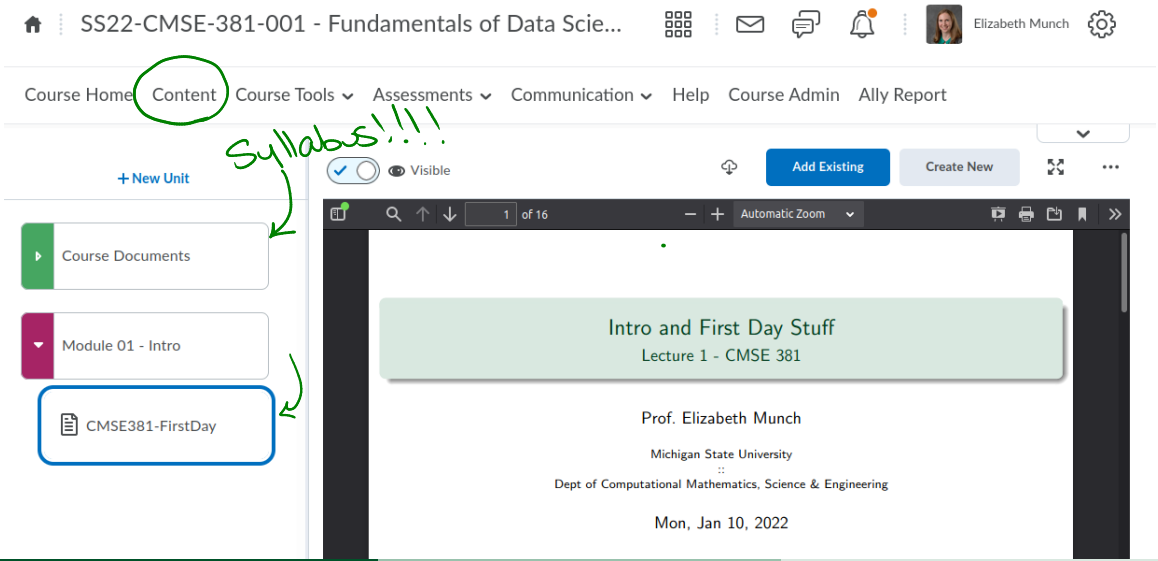
Deep Learning

Zoom and etiquette

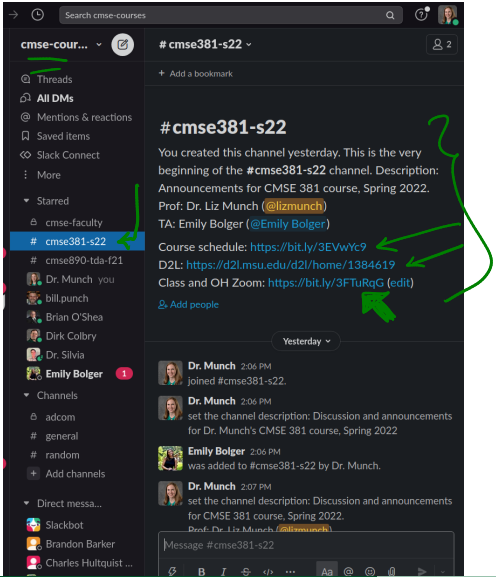
In-Person

- Online until at least February.....
- Please ask questions!
- Cameras
- Kids & Pets

D2L and where to find slides



Slack and where to find announcements/ask questions



Office hours

Same zoom link as class! https://bit.ly/3FTuRqG
Still on Zoom even if we go back in person

Dr. Munch
Mondays and Thursdays 1-2pm

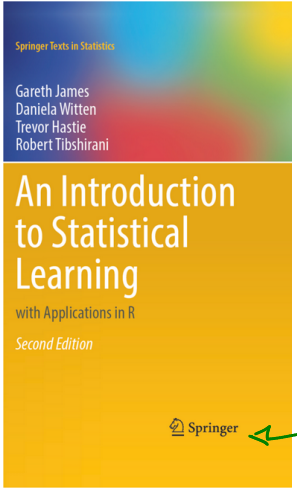
Emily Bolger
Tuesdays 2-4pm
Fridays 12-2pm

Textbook

Free download

https://www.statlearning.com/

Syllabus → 2nd book listed



Lectures & Labs

- Mondays and Wednesdays: Lecture, Some short group work
- Fridays: Labs, group work, graded on participation & submitting end of class survey
  - Bring (ha!) computer
  - Jupyter notebooks
  - Python
  - Drop two lowest grades

Approximate schedule

Up to date version: https://bit.ly/3EVwYc9

	Date	Topic	Reading	Homeworks
M	Jan 10	Intro / First day stuff		
W	Jan 12	What is statistical learning?	2.1	
F	Jan 14	*Lab: Intro to Python*	2.3: Lab	
M	Jan 17	No class - MLK		
W	Jan 19	Assessing Model Accuracy	2.2	
F	Jan 21	Linear Regression	3.1	HW #1 Due
M	Jan 24	More Linear Regression	3.2	
W	Jan 26	Even more linear regression	3.3/3.5	
F	Jan 28	*Lab: Linear Regression*	3.6: Lab	HW #2 Due
M	Jan 31	Intro to classification, Logisitic Regression	4.1, 4.2, 4.3	
W	Feb 2	Generative models for classification	4.4	
F	Feb 4	Lab???	Lab??	HW #3 Due
M	Feb 7	Comparison of classification methods	4.5	
W	Feb 9	Generalized Linear models	4.6	
F	Feb 11	*Lab: Classification Methods*	4.7: Lab	HW #4 Due
M	Feb 14	Resampling methods: CV	5.1	
W	Feb 16	Resampling methods: Bootstrap	5.2	
F	Feb 18	*Lab: Resampling*	*5.3 Lab*	HW #5 Due
M	Feb 21	Subset selection/ Shrinkage	6.1, 6.2	
W	Feb 23	Dimension Reduction	6.3	
F	Feb 25	*Lab: Linear Models & Regularization methods*	*6.5: Lab*	HW #6 Due
M	Feb 28	High dimensions	6.4	
W	Mar 2	Overflow / REVIEW		
F	Mar 4	**MIDTERM**		

M	Mar 7	No class - Spring break		
W	Mar 9			
F	Mar 11			
M	Mar 14	Polynomial & Step Functions	7.1, 7.2	
W	Mar 16	Basis functions, Regression Splines	7.3, 7.4	
F	Mar 18	Lab?	Lab?	HW #7 Due
M	Mar 21	Smoothing Splines	7.5	
W	Mar 23	Generalized Additive Models	7.7	
F	Mar 25	Lab: Non-linear modeling	7.8 Lab	HW #8 Due
M	Mar 28	Decision Trees	8.1	
W	Mar 30	Ensemble methods	8.2	
F	Apr 1		8.3 Lab	HW #9 Due
M	Apr 4	Maximal Margin Classifier, SVC	9.1,9.2	
W	Apr 6	SVM	9.3, 9.4, 9.5	
F	Apr 8	Lab: SVM	9.6	HW #10 Due
M	Apr 11	Single/Multi Layer NN	10.1, 10.2	
W	Apr 13	CNN	10.3	
F	Apr 15	Recurrent NN	10.5	HW #11 Due
M	Apr 18	When to use Deep Learning	10.6	
W	Apr 20	Fitting a NN	10.7, 10.8	
F	Apr 22	*Lab: Deep Learning*	*10.9 Lab*	HW #12 Due
M	Apr 25	Unsupervised Learning & PCA	12.1, 12.2	
W	Apr 27	Clustering	12.4	
F	Apr 29	Lab: PCA & Clustering	12.5 Lab	
	May 4	**FINAL: 5:45-7:45pm**		

Homework

Typed!

Word
Later - overleaf
Hand-drawn figs
are fine

- Weekly homework
- Problems assigned after Monday and Wednesday class
- Homework uploaded to D2L by Midnight Fridays
- Homeworks submitted before midnight Saturday will still be graded with a 5% penalty.
- Homeworks submitted before midnight Sunday will still be graded with a 15% penalty.
- After midnight Sunday, no credit will be given.
- Two dropped homeworks at end of semester.

Midterm and Final

- Midterm: Friday before spring break, March 4
- Final: May 4 5:45 pm

Grade distribution

	Estimated Points
Homeworks	(12 homeworks - 2 lowest grades) $\times$ 20 points $=$ 200
Labs	(12 Labs - 2 lowest grades) $\times$ 10 points $=$ 100
Midterm	100
Final Exam	100
TOTAL:	500

Pandemic specific notes

- This is all weird.....
- Online until at least Feb 1
- Please remember: There are people on the other side of the screen!
- Flexibility and grace
- You need to take ownership of your learning

We are here to help but you need to ask!

Section 1

Intro to class

What is Statistical Learning?

AI

Statistical Learning

- Subfield of statistics
- Emphasizes models and their interpretability, precision, and uncertainty

Machine Learning

- Machine learning has a greater emphasis on large scale applications and prediction accuracy.

Very blurred distinction at this point....

Why should you care?

Data is cheap (or even free), learning how to analyze data is critical.

- Web data, e-commerce (Amazon, JD, Alibaba)
- Car sales (Tesla, Ford, and GM)
- Sports team (MSU, Lions, etc)
- Politics and government

Learning Tools as Black Boxes

Don't know all the little steps

- Need to know what tool to use
- Need to know how to interpret output of the tool
- Don't need to rebuild the entire box from scratch

Example: Email spam

	x_1	x_2	x_3	x_4	x_5	...					
	`george`	`you`	`your`	`hp`	`free`	`hpl`	`!`	`our`	`re`	`edu`	`remove`
`spam`	0.00	2.26	1.38	0.02	0.52	0.01	0.51	0.51	0.13	0.01	0.28
`email`	1.27	1.27	0.44	0.90	0.07	0.43	0.11	0.18	0.42	0.29	0.01

if (`%george` < 0.6) & (`%you` > 1.5) then `spam` else `email`.

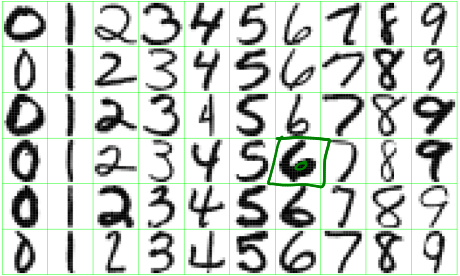
if $(0.2 \cdot$ `%you` $- 0.3 \cdot$ `%george`$) > 0$ then `spam` else `email`.

Classifier

MNIST

Dataset

Handwritten digits



classifier

decide: is this image the number X

$x_i = 0/1$ in pixel i

Supervised learning

MNIST Y = the number

- Outcome measurement Y (also called dependent variable, response, target, label).
- Vector of p predictor measurements X (also called inputs, regressors, covariates, features, independent variables).

  $X_1 \; X_2 \; X_3 \; \cdots \; X_p$
- In the regression problem, Y is quantitative (e.g price, blood pressure).
- In the classification problem, Y takes values in a finite, unordered set (survived/died, digit 0-9, cancer class of tissue sample).

MNIST data

$0, 1, 2, \cdots, 9$

Spam vs not spam

Unsupervised learning

No Y

Yes X

- No outcome variable, just a set of predictors (features) measured on a set of samples.
- Objective is fuzzier: find groups of samples that behave similarly, find features that behave similarly, find linear combinations of features with the most variation.
- Difficult to know how well you are are doing.
- Different from supervised learning but can be useful as a pre-processing step for supervised learning.

Dimensionality reduction

Data Ethics

- Data affects people
- Your decisions on how to analyze and interpret data matter

Next time

- Wednesday: What is statistical learning?
- Friday: Lab, Review of python and jupyter notebooks
- No homework until next week